PUBLIC SOURCES

World of Logs v2

public Jira issues, each flagged for whether its text contains log output

Public Jira Dataset

creation and resolution dates, and duplicate links between issues

Four git histories

Apache Kafka (Java)
MariaDB (C / C++)
MongoDB (C++)
Qt (C++)

four organizations, two tracker platforms

MINE THE BENCHMARK

Join, then filter

keep fixed Bug issues whose text carries logs, link them to 1-3 fix commits by issue key

eligible issues	17,995
no linkable fix commit	-9,134
no source file in the fix	-2,206

Ground truth, from git

the snapshot is the parent of the fix commit; the gold files are the source files it modified

never a file the fix only added

Incident memory pool

every resolved issue with a linked fix, kept per project

2,315 to 19,820 incidents

THE BENCHMARK

6,437
real incidents

each one pairs a report and its logs, the code of that day, and the files the fix later changed

Stratified by difficulty

in **53%** of reports, no file the fix touched is named anywhere

Two evaluation populations

Full evaluation split 4,508
later 70% of each project by fix time; benchmark-wide measurements

Study population 790
200 per project, test-only gold removed; **every** system rescored here

COMPARE, ON ONE POPULATION

one protocol, one temporal firewall

Acc@5

Classic IR localization best of rVSM, BugLocator, BRTracer-style	47.3%
Retrieval + incident memory lexical, deterministic, CPU only	68.6%
A 27B model, given the file tree Agentless-style; re-ranker variant 77.2%	77.8%
Fusion of the two above best system that is not an agent	80.5%
REMEMBER (this paper) same 27B model, given tools	88.8%

No family dominates. Retrieval and the model solve **different** instances; union 86.5%

CHECK THE RESULT

Did it memorise?

asked for each fix from the issue key alone, the model scores **below** a frequency prior

Is it echoing retrieval?

shuffling the candidate order changes **no** result, while its own picks overlap drops 60% to 21%

Is the memory honest?

dropping the temporal firewall inflates the classic similar-report signal by **9% to 43%**

Every number

traces to a result file through a released script, check by check